Oxygen is carried in the blood stream in two forms: bound to hemoglobin and dissolved in the blood. Hemoglobin allows the blood to carry far more oxygen than would be possible by dissolution alone. Although it is more accurate to measure the oxygen content directly using laboratory tests, we can estimate the amount of oxygen carried by the blood if we know the patient's hemoglobin level in g/dL, the PaO_2, and the O_2 saturation. The first step is calculating the amount of oxygen bound to hemoglobin.

Calculating Bound O_2

Calculate the amount of oxygen bound to hemoglobin if the patient's hemoglobin level is 14.5 g/dL and the O_2 saturation is 97%.

Hemoglobin carries 1.34 milliliters of oxygen per g of hemoglobin at 100% saturation. To perform this calculation, we will multiply the hemoglobin level by 1.34 and then multiply this by the O_2 saturation written as a decimal. For example, 97% would be written as 0.97.

Bound O_2 = Pt's Hgb level X 1.34 X O_2 Saturation
Bound O_2 = 14.5 g/dL X 1.34 mL/g X 0.97
Bound O_2 = 18.8 mL/dL

As you see, according to our calculation, a patient with a hemoglobin level of 14.5 g/dL carries 18.8 milliliters of oxygen per deciliter of blood at an O_2 saturation of 97%. We have not yet included the amount of O_2 dissolved in the blood, but first, we will practice with just the amount bound to hemoglobin.

Bound O_2 Calculation Practice Problems

Calculate the amount of oxygen bound to hemoglobin if:

1. The hemoglobin level is 11.1 g/dL and the O_2 saturation is 100%.
2. The hemoglobin level is 13.6 g/dL and the O_2 saturation is 94%.
3. The hemoglobin level is 16.1 g/dL and the O_2 saturation is 90%.
4. The hemoglobin level is 9.8 g/dL and the O_2 saturation is 99%.
5. The hemoglobin level is 14.0 g/dL and the O_2 saturation is 81%.

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Calculating Dissolved O_2

Calculating the amount of oxygen dissolved in the blood is obtained by multiplying the PaO_2 in mmHg by 0.003.

How much O_2 is dissolved in the blood if the PaO_2 is 88 mmHg?

Dissolved O_2 = PaO_2 X 0.003
Dissolved O_2 = 88 X 0.003
Dissolved O_2 = 0.26 mL/dL

It's easy to see from this calculation how hemoglobin vastly increases the amount of oxygen that can be carried in the blood.

Dissolved O_2 Calculation Practice Problems

Calculate the amount of oxygen dissolved in the blood if the PaO_2 is:

1. 78 mmHg
2. 206 mmHg
3. 104 mmHg
4. 464 mmHg
5. 280 mmHg

Now that you are able to calculate the amount of oxygen bound to hemoglobin and the amount of oxygen dissolved in the blood, you can calculate the total amount of oxygen carried in the bloodstream by performing both calculations and adding the answers together.

Calculate the total amount of oxygen carried in the bloodstream if the hemoglobin is 12.8 g/dL, the PaO_2 is 98 mmHg, and the O_2 saturation is 96%.

Bound O_2 = Pt's Hgb X 1.34 mL/g X O_2 Saturation
Bound O_2 = 12.8 g/dL X 1.34 mL/g X 0.96
Bound O_2 = 16.47 mL/dL

Dissolved O_2 = PaO_2 X 0.003
Dissolved O_2 = 98 X 0.003
Dissolved O_2 = 0.29 mL/dL

Total O_2 = Bound O_2 + Dissolved O_2
Total O_2 = 16.47 mL/dL + 0.29 mL/dL
Total O_2 = 16.76 mL/dL

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Total Blood O_2 Calculation Practice Problems

Calculate the total O_2 carried in the bloodstream if:

1. The Hgb is 13.3 g/dL, the O_2 saturation is 92%, and the PaO_2 is 102 mmHg.
2. The Hgb is 15.5 g/dL, the O_2 saturation is 100%, and the PaO_2 is 201 mmHg.
3. The Hgb is 9.0 g/dL, the O_2 saturation is 94%, and the PaO_2 is 88 mmHg.
4. The Hgb is 12.1 g/dL, the O_2 saturation is 96%, and the PaO_2 is 102 mmHg.
5. The Hgb is 14.8 g/dL, the O_2 saturation is 87%, and the PaO_2 is 93 mmHg.
6. The Hgb is 16.7 g/dL, the O_2 saturation is 61%, and the PaO_2 is 43 mmHg.
7. The Hgb is 10.4 g/dL, the O_2 saturation is 77%, and the PaO_2 is 59 mmHg.
8. The Hgb is 15.9 g/dL, the O_2 saturation is 99%, and the PaO_2 is 125 mmHg.
9. The Hgb is 13.7 g/dL, the O_2 saturation is 90%, and the PaO_2 is 76 mmHg.
10. The Hgb is 9.4 g/dL, the O_2 saturation is 100%, and the PaO_2 is 119 mmHg.

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Bound Oxygen Content Answers

(answers are in mL/dL)

1. 14.9
2. 17.1
3. 19.4
4. 13.0
5. 15.2

Dissolved Oxygen Content Answers

(answers are in mL/dL)

1. 0.234
2. 0.618
3. 0.312
4. 1.390
5. 0.840

Total Oxygen Content Answers

(answers are in mL/dL)

1. 16.7
2. 21.4
3. 11.6
4. 15.9
5. 17.6
6. 13.8
7. 10.9
8. 21.5
9. 16.7
10. 13.0

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